

SHALEM BAKTH SINGH BADAMPUDI

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SUMMARY

Robotics Engineer specializing in ROS2 - based autonomous systems, SLAM, and hardware–software integration. Experienced in deploying real-world robotic platforms under competition-grade constraints, including navigation, perception, and manipulation using LiDAR, UR5, and Nav2.

EDUCATION

Bachelor of Technology - ECE
Rajiv Gandhi University Of Knowledge and Technologies ,ONGOLE

Expected 2026
CGPA:8.1/10

WORK EXPERIENCE

Machine Learning Internship (May 2025 - Aug 2025)

IIIT Kottayam

- Explored and implemented advanced Explainable AI (XAI) and fairness techniques for prediction trustworthiness.
- Integrated interpretability tools including LIME, Saliency Maps, and Grad-CAM while conducting bias mitigation via LRP and Integrated Gradients (IG) to deliver verifiable and ethical AI solutions.

Artificial Intelligence Internship (Oct 2024 - Dec 2024)

Infosys Springboard

- Contributed to a Computer Vision project, gaining hands-on experience across the entire AI model development lifecycle. Fine-tuned pre-trained CNN architectures, achieving high accuracy for the task.
- Demonstrated proficiency in model training, validation, and data pre-processing within a team environment.

PROJECTS

Logistic Cobot-Autonomous Multi-Agent Warehouse System (*ROS, Nav2, RViz, MoveIt, Servoing*)

Designed and simulated a collaborative dual-robot system to fully automate inventory sorting and transport within a dynamic warehouse environment.

- Architected the multi-agent system in Gazebo/ROS, creating a synchronized picking/delivery workflow.
- Optimized logistics operations for higher throughput, validating the system’s scalability and robust performance through minimized human intervention and reliable self-guided movement.

Autonomous Warehouse Rover – NXP AIM (3rd Prize Winner) (*ROS, Perception, Control*)

Developed and deployed a ROS2 navigation stack on a physical B3RB rover for the 2025 Grand Finale. Successfully transitioned simulation models to real-world hardware under high-pressure competition constraints.

- **Perception & SLAM:** Integrated Lidar-based SLAM for real-time mapping and obstacle avoidance, and computer vision for object detection, hardening the system against sensor noise and environmental latency.
- **Hardware Integration & Debugging:** Diagnosed and resolved firmware-to-software bottlenecks across NXP-based actuators and sensors, maintaining system stability during high-pressure live demonstrations

Krishi coBot: Autonomous Precision Agriculture – eYRC 2025-26 (1st Place) (*ROS, Lidar, OpenCV*)

Developing a mobile manipulator system featuring an **eBot rover** and a **UR5 robotic arm** for autonomous farming and sorting. This project focuses on complex navigation logic and real-time perception for precision agriculture.

- **Manipulation & Navigation:** Developing a Python-based waypoint navigator in **ROS 2** and integrating **Servoing** for the UR5 arm to perform precise ”pick-and-place” tasks on target coordinates.
- **Perception & Debugging:** Utilizing **OpenCV** for fruit detection to guide the UR5 arm and **Lidar-based** shape detection for the eBot rover while resolving Gazebo environment conflicts.

SKILLS

Robotics & Systems: ROS2, SLAM, Nav2, MoveIt, Gazebo, UR5, LiDAR, OpenCV
AI/ML/Data: Machine Learning, Deep Learning, NLP, SQL
Programming: Python, C, C++, JavaScript, React, HTML, CSS
Tools/Frameworks: Github, VS Code, Jupyter Notebook, Firebase, MS Office
Soft Skills: Leadership, Communication, Adaptability, Teamwork

ACHIEVEMENTS

- 3rd Prize Winner in Grand finale of NXP AIM Robotics Competition by NXP Semiconductors - 2025
- Regional Finalist Winner in NXP AIM Robotics competition by NXP Semiconductors - 2025
- Top 1% in NPTEL Machine Learning
- Reached to task 4 in Logistic Cobot theme of e-yantra Robotics Competition by IIT Bombay – 2024
- Advanced to Stage-2 in NXP AIM Robotics competition by NXP semiconductors- 2024

CERTIFICATIONS & EXTRACURRICULARS

Certifications: Leetcode DSA in Python (Udemy), SQL (HackerRank), Principles of Generative AI (Infosys)
Leadership: Led multiple robotics competitions as a team, including e-Yantra and NXP Semiconductors challenges, managing end-to-end planning, design, and implementation.